

CSMFAB000000000113-Vol-28

Prototype Build Sequence and Hobbyist Test Protocol: Step-by-Step Guide

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1. Minimum Viable Prototype (MVP): Scale 1 Nano Launch

Goal: Launch a 50 g, 160 mm ZrB₂-SiC spinning spear from 10 m depth using 2 Smart Ropes and 2 paddleboards, achieving 300+ m/s exit velocity and >3 km apogee.

Estimated build time: 8-12 weeks for a 2-person team **Estimated cost:** \$400-\$600 total **Legal status:** 3.5 km apogee < 18,000 ft → below Class E airspace; no FAA notification required. File voluntary NOTMAR if >1 nm offshore.

Phase 1 — Weeks 1-3: Spear Fabrication

Week 1: Acquire ZrB₂ and SiC powders: - ZrB₂: 50g from American Elements (99.5%, -325 mesh): \$42 - SiC: 10g from Alpha Aesar (98%): \$8 - KNbO₃-BaTiO₃ discs (4×): from TDK 2025, \$3.60 each = \$14.40

Tape-cast ZrB₂-SiC green tape per CSMFAB000000000001 protocol. Press into 25mm diameter × 160mm cylinder mold (split steel mold, available from hobby ceramics supplier: \$45).

Week 2: Fire in borrowed university ceramics kiln: Flash sintering if available; conventional SPS otherwise. Dimensional verification: OD 20mm ± 0.2mm, straightness < 0.5mm over 160mm length.

Week 3: Screw blade fabrication: - Cast 3 helical blade segments from ZrB₂-SiC green tape, arc-shaped - Bond to spear body with ZrB₂-SiC sinterable adhesive (CSM Materials Study Part I protocol) - Install KNbO₃-BaTiO₃ flutter patches (4×) with UV epoxy

Phase 2 — Weeks 4-6: Smart Rope Fabrication

Week 4: Acquire Smart Rope components: - UHMWPE Dyneema SK75: 25m × 2 ropes = 50m, 3mm braid: \$12.50 (DSM wholesale) - ZrB₂-SiC segment cubes: 16 per rope × 2 ropes = 32 cubes: machine from sintered blank \$0.87 each = \$27.84 - LoRa RAK3172 modules × 34 (including 2 per rope + 2 master + 2 spear): \$6.50 each = \$221

Week 5: Rope assembly: - Thread UHMWPE through 16 ZrB₂-SiC cubes per 10m rope - Install KNbO₃-BaTiO₃ actuator in each cube - Apply marine-grade viton O-ring seals on each bayonet joint - Hydrostatic test at 15 bar for 1 hour

Week 6: Electronics integration: - LoRa firmware: download open-source LoRa ranging firmware (GitHub: LoRa-Alliance/LoRaWAN-lib) - Timing synchronization: GPS-disciplined crystal oscillators from GPSDXO (\$45 each × 4 = \$180) - PMMA optical fiber trigger cable: 6m × \$0.25 = \$1.50 per rope

Phase 3 — Weeks 7-8: Vessel Preparation and Dry Runs

- Acquire 2 paddleboards (or borrow): \$0-\$400
- Install BFRP rope mount frame on each paddleboard: \$35 materials
- Dry-run rope deployment in swimming pool: verify timing synchronization < 0.2 ms
- Dry-run spear drop to 3m depth in pool: verify positive buoyancy, spin-up motor function

Phase 4 — Weeks 9-10: Offshore Testing

Test 1 — Static rope pull (no launch): - Deploy spear to 5m depth - Rope pull test at 10% max force - Verify synchronized actuation via accelerometer telemetry

Test 2 — Shallow water launch: - Deploy spear to 3m depth - Full rope actuation - Measure exit velocity via high-speed camera (1000 fps): target >50 m/s - Document trajectory arc

Test 3 — Full depth launch: - Deploy spear to 10m depth - Full rope actuation from 2 vessels - Target exit velocity: 250-300 m/s - Track apogee with radar altimeter or optical theodolite

2. Data Collection Protocol

Record during each test: - Video: 4K main + 240fps super-slow for exit transition - LoRa telemetry: acceleration, rotation, pressure vs time - Radar track: if available (or theodolite manual tracking) - Weather: sea state, wind, temperature, water depth

Key metrics: - Water-exit velocity (from video frame analysis or radar) - Trajectory angle vs predicted - Rope synchronization timing (from telemetry) - Spear spin rate at exit (from gyro telemetry)

Citations (Vol 28)

- AeroTech hobbyist rocketry safety guidelines, 2024
- CSMFAB0000000000001 V2.0, ZrB2-SiC fabrication protocol, CSM 2026
- CSMFAB00000000000022 V2.0, Smart Rope manufacturing, CSM 2026
- FAA Advisory Circular AC 101-1, “Moored Balloons and Unmanned Rockets,” 2022
- USCG Navigation Rules, International-Inland, 2022

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